

Example 1: Fetch data with R

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Setup R

```
1 # install.packages("rdbnomics")
2 library(quarto) # for compiling Quarto presentations and documents
3 library(rdbnomics) # for compiling Quarto presentations and documents
4 library(tidyverse) # dplyr, ggplot2, readr, etc.
5 library(here) # path settings
6 library(plotly) # for interactive plots
7 library(countrycode) # for country codes
```

Interfaces to Official Statistics

- Packages or set of classes and methods to read data and metadata documents through exchange frameworks
 - Use [R](#) (or [Python](#)) packages to read data from APIs, databases, and web pages
 - Individual packages:
 - [eurostat](#): Access data from Eurostat
 - [OECD](#): Access data from the OECD API
 - General-purpose packages:
 - [rdbnomics](#): Unified access to many economic databases (e.g. ECB, Eurostat, IMF, World Bank)
 - [pxweb](#): R package developed by Statistics Sweden to disseminate public statistics

DBnomics

- **DBnomics** is a database of databases
 - free platform to aggregate publicly-available economic data provided by national and international statistical institutions, but also by researchers and private companies
 - Unified interface to access data from many sources
 - Harmonized data formats and metadata
 - Data series are available upon release by the provider
 - Each revision is archived to build a real-time database



Example: Fetch Unemployment Data

- Assume we know exactly the series ID we want to fetch ¹
 - Example data: Unemployment rate, ILO definition, total, Estonia, from Eurostat

```
1 unemp <- rdb(ids = "Eurostat/ei_lmhr_m/M.PC_ACT.SA.LM-UN-T-TOT.EE") # fetch data
```

```
1 glimpse(unemp)
```

Rows: 296

Columns: 22

```
$ _@frequency`  
<chr> "monthly", "monthly", "mo...  
"monthly", "monthly", "mo..."
```

```
$ dataset code
<chr> "ei lmhr m", "ei lmhr m", "ei lmhr m..."
```

```
$ dataset name  
<chr> "Unemployment rate (%)" - monthly dat...
```

```
$ freq
```

```
$ geo
<chr> "EE", "EE", "EE", "EE", "EE", ...
```

```
$`Geopolitical entity (reporting)`<chr> "Estonia", "Estonia", "Es..."
```

```
$ indexed_at
<dtm> 2024-10-31 15:26:51, 2024-10-31 15:...
```

```
$ indic
<chr> "LM-UN-T-TOT", "LM-UN-T-TOT", "LM-UN-...
```

```
$ Indicator
<chr> "Unemployment according to ILO defin..."
```

```
$ observations attributes
<chr> "OBS FLAG", "OBS FLAG...", "OBS FLAG..."
```

```
$ original_period
<chr> "2000-02", "2000-03", "2000-04", "20..."
```

```
$ original value
<chr> "14.9", "14.2", "14.5", "13.9", "14..."
```

```
<date> 2000-02-01, 2000-03-01, 2000-04-01,...
$ period
```

```
$ provider code
<chr> "Eurostat", "Eurostat", "Eurostat", ...
),

```

```
$ s_adj
<chr> "SA", "SA", "SA", "SA", "SA", ...
```

```
$ `Seasonal adjustment`  
+ `Seasonally adjusted data, not calen...`  
  
<chr> "Seasonally adjusted data, not calen..."
```

```
$ series code
<chr> "M.PC ACT.SA.LM-UN-T-TOT.EE", "M.PC ...
```

```
$ series_name
<chr> "Monthly - Percentage of population "
```

Work with the dataframe

We can directly work with the dataframe `unemp` as it is already in a tidy format

```
1 # Extract source and series ID from the metadata
2 (source_name <- unique(unemp$dataset_code))
```

```
[1] "ei_lmhr_m"
```

```
1 (provider_code <- unique(unemp$provider_code))
```

```
[1] "Eurostat"
```

```
1 (country_name <- unique(unemp$`Geopolitical entity (reporting)`))
```

```
[1] "Estonia"
```

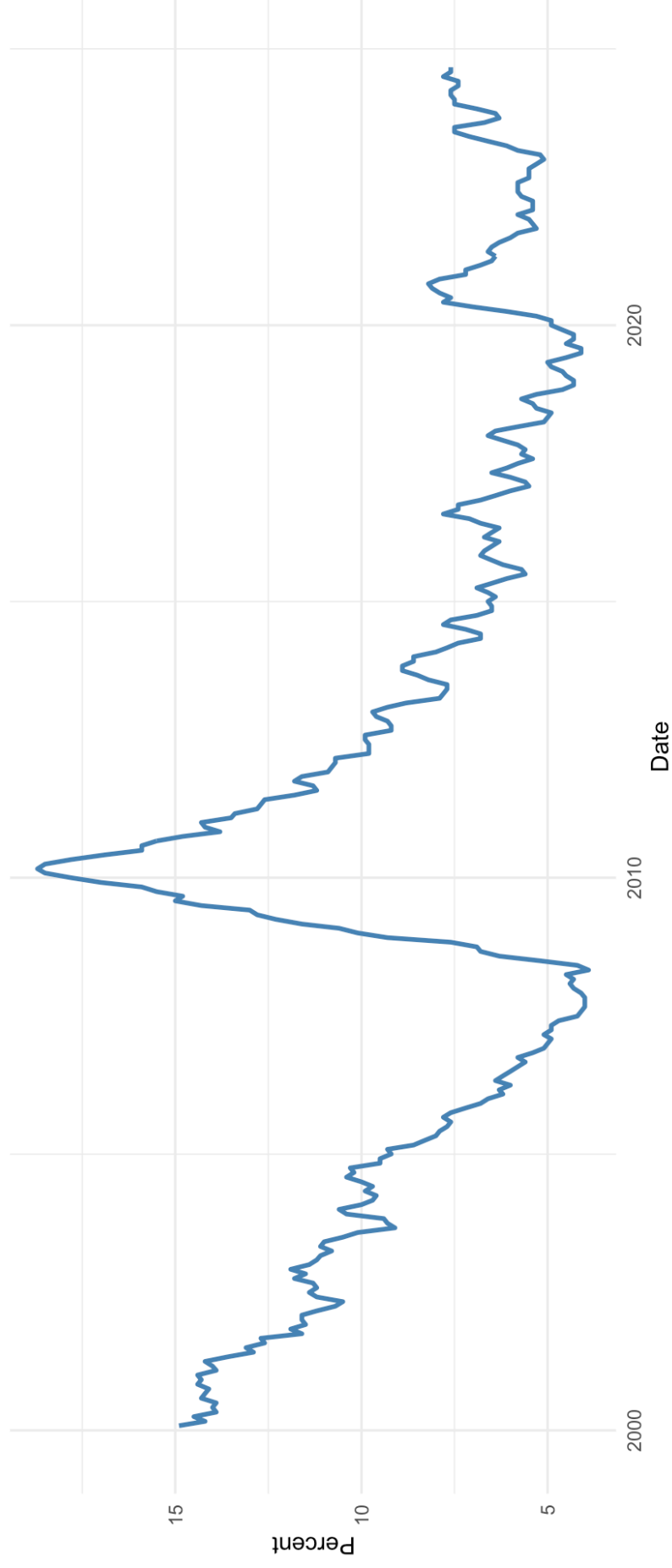
```
1 (series_id <- unique(unemp$series_code))
```

```
[1] "M.PC_ACT.SA.LM-UN-T-TOT.EE"
```



```
1 # Plot the data
2 p1 <- ggplot(unemp, aes(x = period, y = value)) +
3   geom_line(color = "steelblue", linewidth = 1) +
4   labs(
5     title = paste("Unemployment Rate in ", country_name),
6     subtitle = paste("Monthly, seasonally adjusted -", provider_code),
7     x = "Date", y = "Percent",
8     caption = paste("Source:", provider_code, " | Dataset:", source_name, " | ID:", series_id)
9   ) +
10  theme_minimal()
11 p1
```

Unemployment Rate in Estonia
Monthly, seasonally adjusted — Eurostat



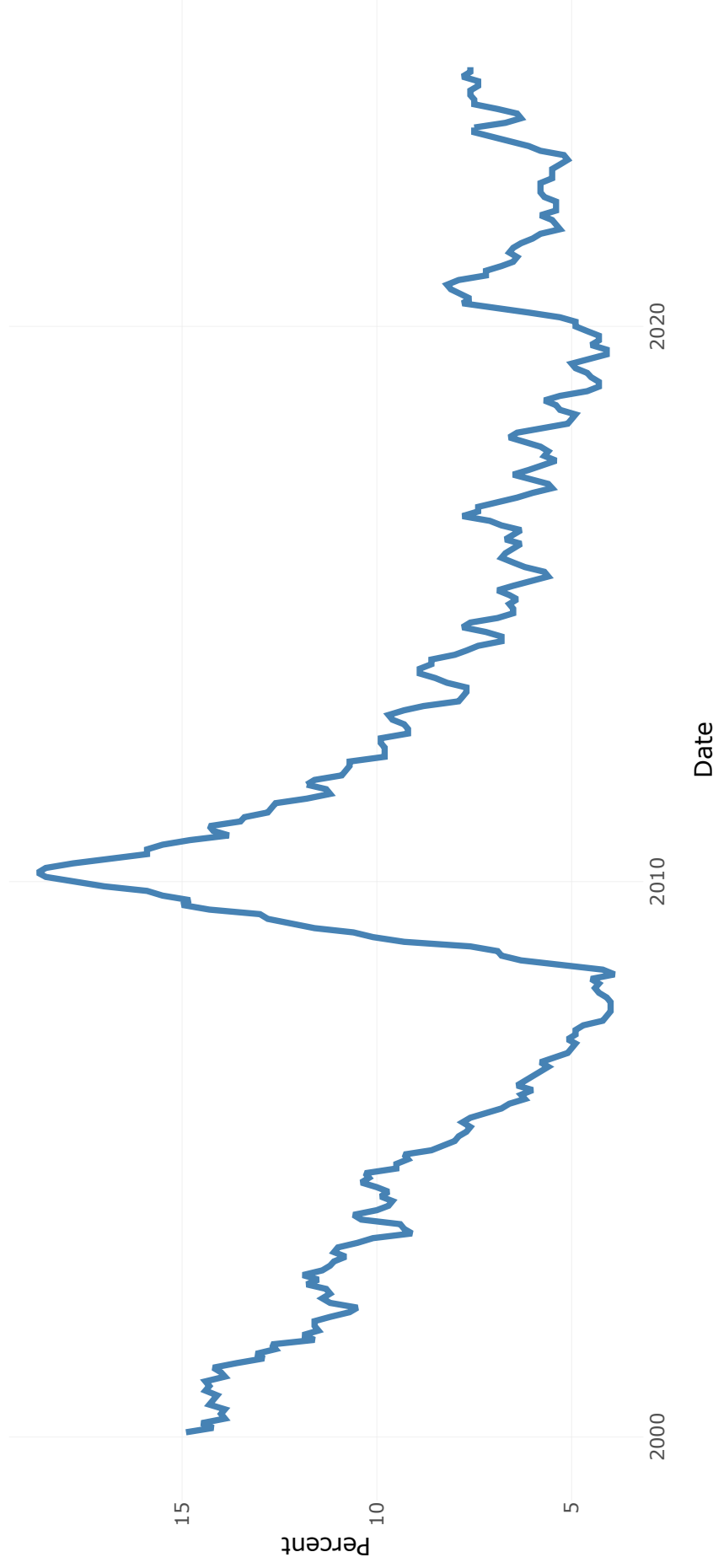
Source: Eurostat | Dataset: ej_lmhr_m | ID: M.PC_ACT.SA.LM-UN-T-TOT.EE

Interactive plot

When working with dynamic documents, I can also create interactive plots using the `plotly` package.

```
1 ggplotly(p1)
```

Unemployment Rate in Estonia



But I Need a Plot That Matches Eesti Pank's Corporate Design

- Use **ChatGPT** to extract the official color palette and font settings from Eesti Pank's design templates (e.g., *Tööturu ülevaade_2025.xlsx*)
- Define a custom **ggplot2** theme that matches the corporate look - `theme_ep()`

```
1 library(ggplot2)
2
3 # Color scale
4 scale_color_ep <- function(...) {
5   scale_color_manual(
6     values = c(
7       "#0079BC", # Accent 1
8       "#D6E1F2", # Accent 2
9       "#FF8C42", # Accent 3
10      "#8ACC61", # Accent 4
11      "#E85C70", # Accent 5
12      "#E091D4", # Accent 6
13    ),
14    ...
15  )
16 }
17
18 # Fill scale
19 scale_fill_ep <- function(...) {
20   scale_fill_manual(
```

```
21 values = c(  
22   "#0079BC",  
23   "#D6E1F2",  
24   "#FF8C42",  
25   "#8ACC61".
```

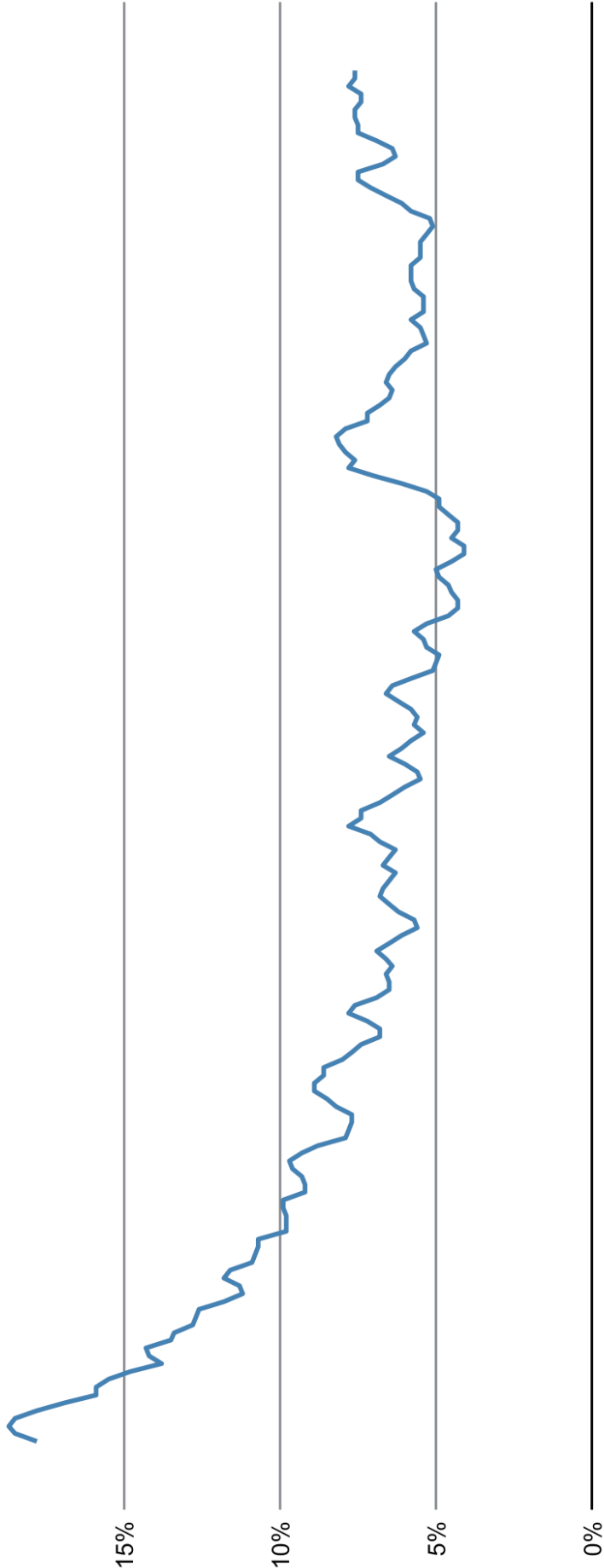
```

1 unemp |>
2   filter(period >= "2010-01-01") |>
3   ggplot(aes(x = period, y = value)) +
4   geom_line(color = "steelblue", linewidth = 1) +
5   geom_hline(yintercept = 0, color = "#000", linewidth = 0.5) +
6   scale_x_date(
7     date_breaks = "1 year",
8     date_labels = "%Y"
9   ) +
10  scale_y_continuous(
11    labels = scales::label_number(suffix = "%"),
12    name = NULL
13  ) +
14  labs(
15    title = paste("Unemployment Rate in", country_name),
16    subtitle = paste("Monthly, seasonally adjusted"),
17    x = NULL,
18    caption = paste(
19      "Source:",
20      provider_code,
21      "| Dataset:",
22      source_name,
23      "| ID:",
24      series_id
25    )

```

Unemployment Rate in Estonia

Monthly, seasonally adjusted



2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024 2025
Source: Eurostat | Dataset: ei_lmhr_m | ID: M.PC_ACT.SA.LM-UN-T-TOT.EE

But I Need to Store the data as an Excel File

```
1 library(openxlsx)
2
3 wb <- createWorkbook()
4 addWorksheet(wb, "Unemployment Data")
5 writeData(wb, "Unemployment Data", unemp)
6 saveWorkbook(wb, "teaching_materials/unemp_data.xlsx", overwrite = TRUE)
```

- Exporting editable figures: Advanced topic!
 - Use {officer} + {rvg} to embed editable vector graphics in Excel or use {mschart} to create editable charts for Powerpoint or Word

How do we find the series ID/mask/dimensions?

- Go to the [DBnomics website](#)
 - Search directly for a series or pick a provider
 - Search for the [data](#) you want (dataset_code)
 - Click on the [series](#) (series_code)
 - Copy the series ID from the URL

Fetch two (or more) series at once

- Example data: Balance of Payments (BOP) for France and Germany from the [IMF](#) for Current Account, Total, Net, Euros, Millions, Annual

Option A

Option B:

Option C:

```
1 # by ID
2 bop <- rdb(ids = c("IMF/BOP/A.FR.BCA_BP6_EUR", "IMF/BOP/A.DE.BCA_BP6_EUR"))
3 bop %>% count(`Reference Area`)
```

	Reference Area	n
1:	France	15
2:	Germany	26

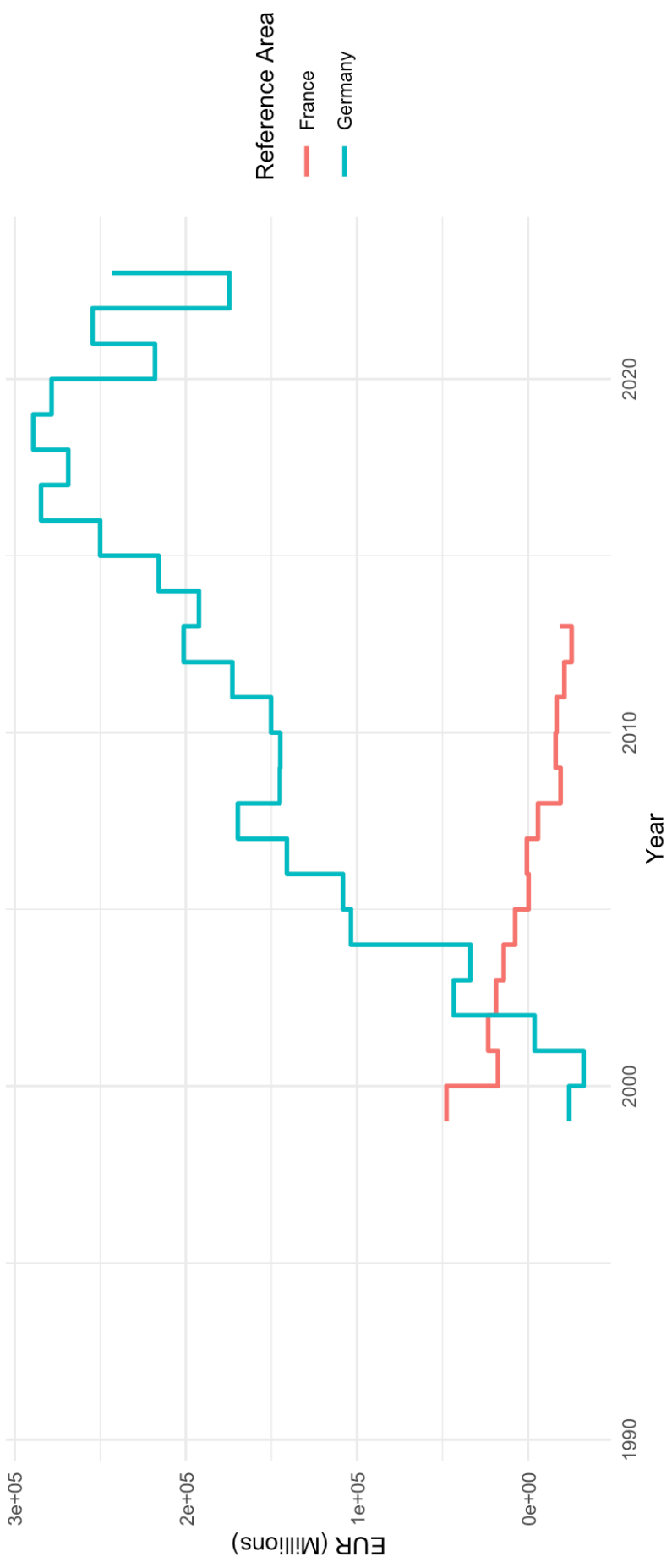
```

1 # Line plot with color by country
2 p2 <- ggplot(bop, aes(x = period, y = value, color = `Reference Area`)) +
3   geom_step(linewidth = 1) +
4   labs(
5     title = "Balance of Payments (BCA, EUR)",
6     subtitle = "France vs Germany – Annual",
7     x = "Year",
8     y = "EUR (Millions)",
9     caption = "Source: IMF / DBnomics"
10  ) +
11    theme_minimal()
12 p2

```

Balance of Payments (BCA, EUR)

France vs Germany — Annual



Source: IMF / DBnomics

Wildcarding dimensions

- To fetch **all** for a dimension (e.g. countries), just **remove** the value from that position
 - Example: remove **"EE"** to fetch all countries (**REF_AREA**)
- Example data: Consumer confidence index from the OECD, downloading data on all countries

⚠ This can take a while

```
1 # cci_ee <- rdb("OECD", "DSD_HHDASH@DF_HHDASH_CTRY", "Q.EST.CCICP.IX")
2 cci <- rdb("OECD", "DSD_HHDASH@DF_HHDASH_CTRY", "Q..CCICP.IX")
3 # cci <- rdb("OECD", "DSD_HHDASH@DF_HHDASH_CTRY", "..CCICP.IX") # also wildcard the frequency
4 unique(cci$REF_AREA)
```

[1]	"AUS"	"AUT"	"BEL"	"CHE"	"CHL"	"COL"
[7]	"CRI"	"CZE"	"DEU"	"DNK"	"EA"	"ESP"
[13]	"EST"	"EU27_2020"	"FIN"	"FRA"	"G7"	"GBR"
[19]	"GRC"	"HUN"	"IRL"	"ISR"	"ITA"	"JPN"
[25]	"KOR"	"LTU"	"LUX"	"LVA"	"MEX"	"NLD"
[31]	"NZL"	"OECD"	"POL"	"PRT"	"SVK"	"SVN"
[37]	"SWE"	"TUR"	"USA"			

```

1 cci_eu <- cci %>%
2   filter(
3     REF_AREA %in% c("EU27_2020", "AUT", "EST", "FIN", "DEU", "LVA", "LTU", "POL", "SWE")) |>
4   mutate(
5     REF_AREA = case_when(
6       REF_AREA == "EU27_2020" ~ "EU27",
7       TRUE ~ countrycode(REF_AREA, origin = "iso3c", destination = "iso2c")
8     )
9   )
10  unique(cci_eu$`Reference area`)

```

```

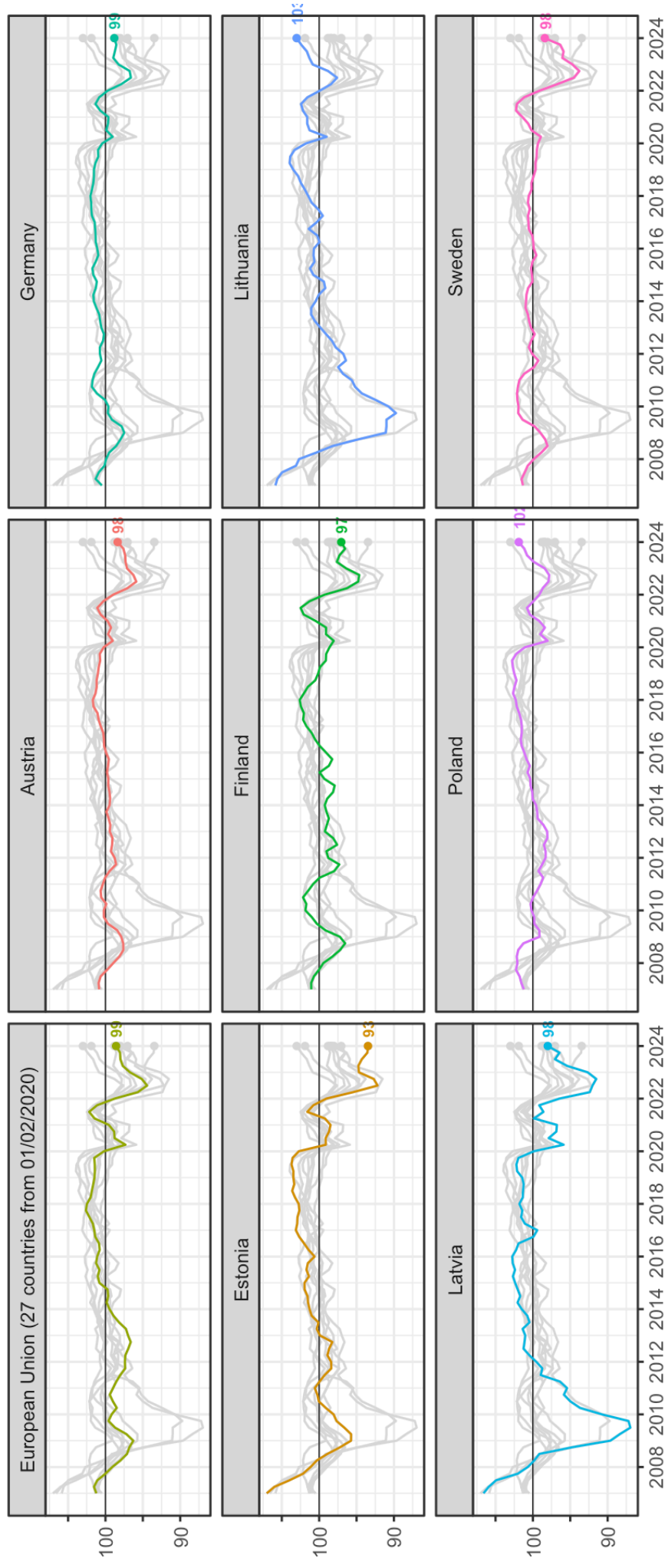
[1] "Austria"
[2] "Germany"
[3] "Estonia"
[4] "European Union (27 countries from 01/02/2020)"
[5] "Finland"
[6] "Lithuania"
[7] "Latvia"
[8] "Poland"
[9] "Sweden"

[1] "Austria"
[2] "Germany"
[3] "Estonia"
[4] "European Union (27 countries from 01/02/2020)"
[5] "Finland"
[6] "Lithuania"
[7] "Latvia"
[8] "Poland"
[9] "Sweden"

```

Consumer Confidence – Index

Selected European Countries & EU27



Source: OECD / DBnomics

End of session